

Overuse Injury Risk Factors

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Intrinsic Factors – Originate within

- **Malalignment**
 - Check patient for foot posture, patella heights, hip angle, bow leggedness
- **Anatomical irregularities**
 - Leg length discrepancy, muscle imbalance, weakness
- **Flexibility**
 - Generalized muscle tightness, focal areas of thickening, restricted ROM
- **Other**
 - Genetic, endocrine, metabolic

Categories of Overuse Injuries

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- Bone Stress
 - Stress reaction / Stress fracture
- Muscle Overuse
 - Focal tissue thickening and fibrosis
 - Chronic compartment syndrome
 - Muscle soreness “DOMS”
- Tendinopathy



Skeletal Injury

Tibial stress fracture – Makes up ~50% of all diagnosed stress fractures

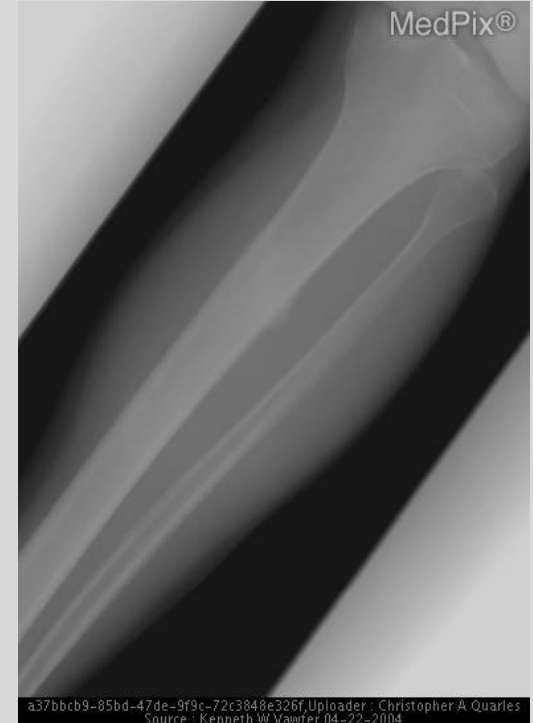
- Typically secondary to running
- Two possible locations
 1. Between the middle and distal third of the tibia along the posteromedial border – Associated with *running*
 2. Along the middle third of the anterior cortex – Associated with *Jumping* activities (basketball, dancing, gymnastics)

CURRENT Diagnosis & Treatment Family Medicine Chapter 39: Common Upper & Lower Extremity Fractures

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Skeletal Injury

Extrinsic Risk Factors of Stress Reaction / Fracture

- Changes in exercise regimen;
 - Runners who increase mileage too quickly
 - Increased intensity
 - Introduction of a new activity
 - Change in training surface
 - Change in training equipment

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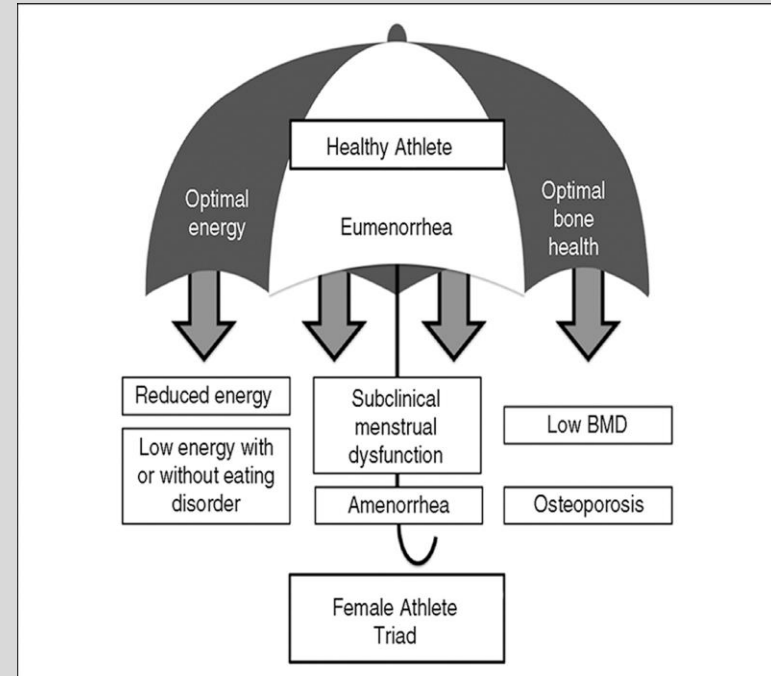
Skeletal Injury

- Intrinsic Risk Factors of Stress Reaction/Fracture
 - Low bone density
 - Low BMI
 - Dietary deficiency
 - Hormonal imbalance
 - Sleep Deprivation
- Increased risk in women
 - Female athletic triad: Amenorrhea, Disordered eating and Osteoporosis

Source: JAAOS - Journal of the American Academy of Orthopaedic Surgeons 23(7):424-432, July 2015.

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Skeletal Injury

- Clinical Presentation:
 - Typically occurs in the lower extremity due to repetitive weight bearing
 - Athletes, specifically runners are at highest risk for stress reaction / fracture
 - **Crescendo effect:** Patients report days to weeks of dull but growing pain eventually leading to inability to perform an activity
 - Initially, pain decreases with rest, but as time progresses, pain returns with less and less activity
 - Finally, pain may be present with very minimal activity and bothersome at night

Skeletal Injury Diagnosis

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- Clinical – based on patient presentation
- Imaging – Confirmation
 - Plain radiograph: Low sensitivity but high specificity. May only be positive after symptoms present 4-6 weeks
 - Positive signs include;
 - Periosteal reaction, Intramedullary sclerosis, Obvious fracture line
 - Technetium Bone Scan: High diagnostic power for stress fracture. Can be positive within 48 hours of symptom onset
 - **MRI- best test to confirm diagnosis**

Stress Fracture Treatment

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General Principles

- #1 if it hurts, DO NOT DO IT
- Typically, 6 weeks of *relative* rest are required
- Bracing has been shown to decrease time to return to competition by 20 days
- Avoid NSAIDs – may hinder bone healing
- Maintain fitness with cross training throughout rest period
- Correct identified precipitating factors
 - Muscle imbalance (strength training), mechanical abnormalities (orthotics), dietary planning, and training modification

Muscle overuse

- Typically secondary to muscle imbalances
- Imbalances lead to changes in muscle length and strength between agonists and antagonists – effecting overall muscle function
- This leads to overall functional and biomechanical problems leading to injury
- Muscle injuries include
 - Focal tissue thickening and fibrosis
 - Chronic compartment syndrome
 - Muscle Soreness

Muscle Overuse

Focal tissue thickening and fibrosis

- Repetitive micro trauma of overuse
 - Damage of muscle fibers
- Damage leads to the formation of **adhesions** between muscle and **cross linkages** in the fascia
- These cross linkages and adhesions may be palpated as focal areas of tissue thickening
 - Taut, thickened band arranged in the direction of stress or as large area of increased muscle tone

Focal tissue thickening and fibrosis

- Focal areas of thickening may be responsible for
 - Pain
 - Predisposition to injury of tendons due to a lack of tissue elasticity under stretch or eccentric load
 - Decreased ability for muscle to contract and relax rapidly
- Treatment for focal tissue thickening and fibrosis
 - Soft tissue therapy (OMT)
 - Strengthening
 - Stretching

Chronic Compartment Syndrome

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Muscle overuse - Intermittent and reversible pathologic elevation of compartment pressure following exertion

- Most common in the lower leg
 - Can occur in forearm (tennis, rock climbing, weight lifting)
- Compartments
 - Created by relatively inelastic fascial sheaths of collagenous tissue

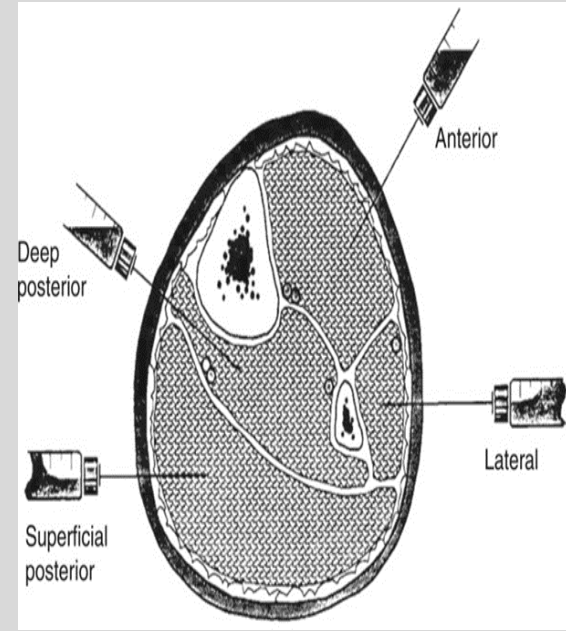


Chronic Compartment Syndrome

- **Clinically:** The main symptom is pain and pressure that commences with activity and dissipates with rest
- **Measurement of compartment pressures**
 - Can be done at rest, or during pain provoking exercise
 - Catheter placed into muscle compartment suspected to cause pain
 - First exercise muscle until symptom reproduction.
 - Next measure compartment pressure for 5 minutes.

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Tendinopathy

- Clinical presentation
 - Pain with use or in the morning localized to tendon
 - Typically pain free with rest
 - **Typical pain pattern:** Pain begins with exercise, but decreases as patient continues to warm up, then returns as patient cools down
 - **Physical exam:** Local tenderness and thickening of the associated tendon

Tendon Overuse

- **Diagnosis and Treatment**
- **Diagnosis:**
 - Clinical – pattern of pain
 - Imaging - Ultrasound hypoechogenicity, increased sized and thickness of tendon
- **Treatment:**
 - Conservative: Physical therapy, stretching, activity modification, change of footwear, injections (avoid steroids due to tendon rupture risk)
 - Operative: Tendon augmentation or tendon transfer (FHL transfer to Achilles)

A 44-year-old patient presents to the clinic complaining of intermittent right forearm pain and pressure of three months duration. Pain is achy and tight with occasional numbness that is present when they plays tennis but subsides immediately after. What is the most likely diagnosis?

- A. Stress fracture
- B. Tennis elbow
- C. DOMS
- D. Chronic compartment syndrome
- E. Muscle strain

A 44-year-old patient presents to the clinic complaining of intermittent right forearm pain and pressure of three months duration. Pain is achy and tight with occasional numbness that is present when they plays tennis but subsides immediately after. Which of the following is the best initial treatment?

- A. Fasciotomy
- B. Fasciectomy
- C. Compartment pressure measurement
- D. Casting
- E. Physical Therapy

A 22-year-old male presents to the clinic complaining of right shoulder pain radiating to the anterior upper arm. On physical exam when the right shoulder is flexed and externally rotated with thumb pointed up and the patient is resisting downward pressure the pain is reproduced. What would you expect to find on histopathology?

- A. Lymphocytes
- B. Neovascularization
- C. Increased osteoclast activity
- D. Organized tendon
- E. Muscle necrosis

A 22-year-old male presents to the clinic complaining of right shoulder pain radiating to the anterior upper arm. On physical exam when the right shoulder is flexed and externally rotated with thumbs pointed up and the patient is resisting downward pressure the pain is reproduced. Which of the following treatments should be avoided?

- A. Relative rest
- B. Physical therapy
- C. OMM
- D. Corticosteroid injection